



Assessment and Redesign of ESI Data Center: Toward a Modern Scalable Architecture

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Itroduction:

A data center is the core infrastructure that hosts and delivers critical services such as applications, storage, computing resources, and network connectivity.

At ESI, the current data center reliably supports existing services, but it is not designed to meet the growing demands of scalability, high-performance computing (HPC), and AI-driven workloads.

So our objectives from this work is to:

- Evaluate ESI current infrastructure.
- Identify its limitations.
- Propose modern leaf-spine logical & physical architectures.
- Improve performance, scalability and security.

Study of Existing Data Center: Current Architecture

ESI operates an on-premises data center designed around a traditional Core / Distribution / Access architecture.

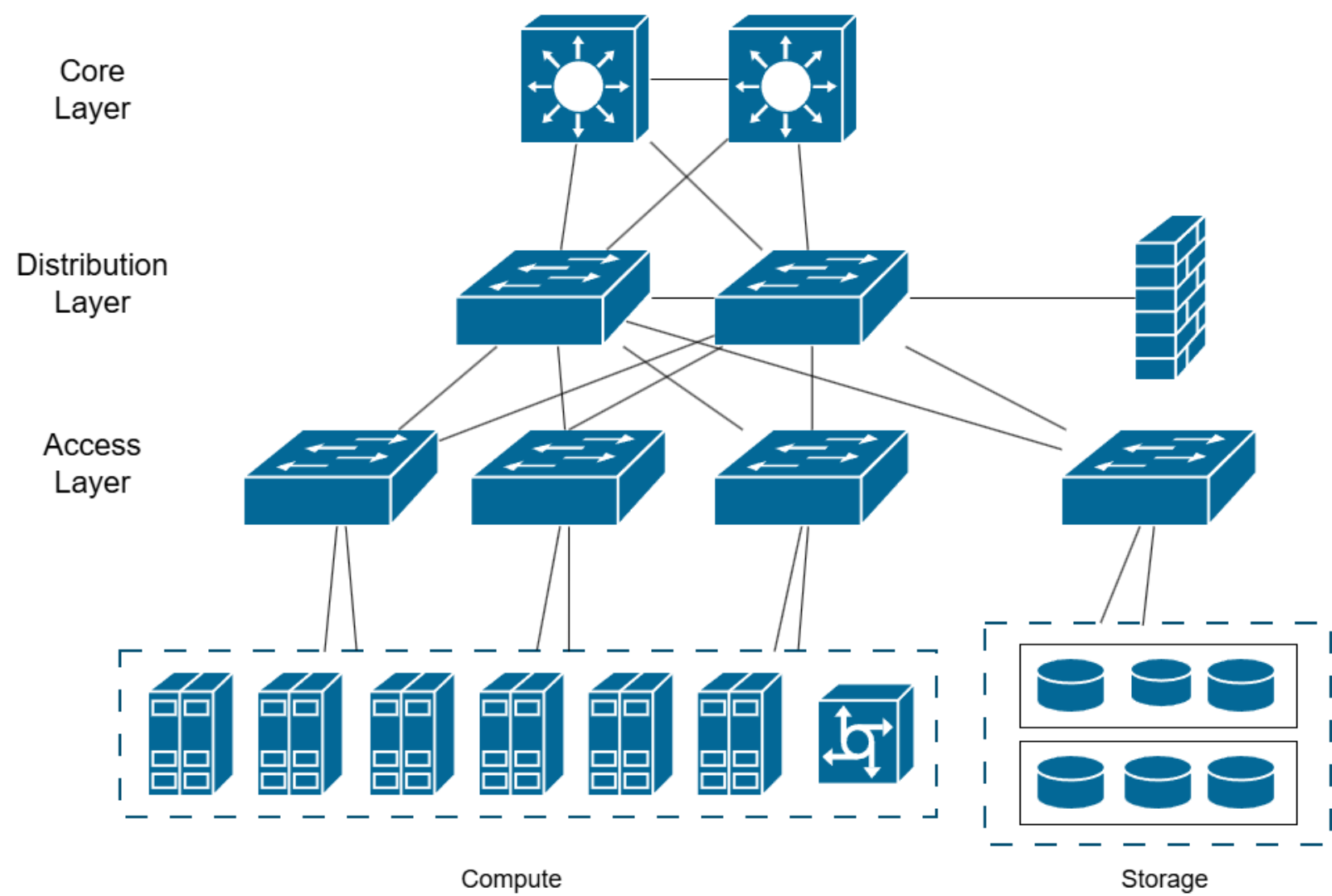


Figure 1. High-Level Conceptual Topology for Current ESI Architecture

We can summarize the architecture in these two tables:

Networking	Compute
3-tier architecture (Core / Distribution / Access) Layer 3 switching VLAN segmentation (per building & user type) Dual Internet: Fiber + Satellite (backup) Internet Bandwidth ≈ 1 Gbps	~40 Physical Servers (HP DL380 Gen) VMware ESXi virtualization Intel Xeon CPUs Max 128 GB RAM/server 1 GPU

Table 1. Networking and compute infrastructure.

Storage	Security and Monitoring
2 NAS systems Redundancy for critical services No performance monitoring	Sophos software firewall 2 Physical firewalls (Baccalaureate - BAC only) Local networking monitoring system

Table 2. Storage and security/monitoring infrastructure.

ESI Services:

The ESI services are categorized as follows:

Category	Services Provided
Educational & Academic	Moodle platform, Talent platform...
Cloud & Compute	Web services, virtual machines (Linux, teaching environments)
Administrative	Internal applications, institutional websites
National Services	BAC orientation platform

Table 3. Overview of provided ESI services.

The analysis of the existing infrastructure highlights several technical and operational constraints. These limitations impact performance, scalability, and the ability to support future needs.

Limitations:

- Outdated switching and server infrastructure limits performance and **Scalability**.
- Very limited GPU resources, not suitable for AI/ML workloads.
- No dedicated HPC infrastructure.
- Storage system not optimized for large-scale or research datasets.
- Lack of centralized monitoring for network, system, and security management.
- Incomplete backup and disaster recovery strategy.
- Weak environmental monitoring (e.g., temperature, system health).

Analysis & Requirements:

The current ESI data center supports essential academic and administrative services with acceptable stability. However, it is primarily designed for traditional workloads and shows clear limitations, especially that ESI will provide new services:

Research HPC & AI

Scientific simulations, HPC workloads, AI/ML & deep learning, large dataset processing.

Cloud

Student project hosting, VMs (virtual machines) provision.

So the fully requirements are:

- Support and ensure availability of existing academic, administrative, and institutional services.
- Enable high-performance computing for AI, machine learning, and scientific research workloads.
- Provide **scalable**, high-availability infrastructure with centralized monitoring and improved resource management.
- Ensure strong security, backup, and environmental monitoring for reliable and continuous operation.

Proposed Solution Overview:

As we saw in our previous theory study of leaf-spine architecture, this last is adopted as it provides a modern network design that meets ESI's needs for performance, scalability, and future growth.

So we based our proposed solution above this architecture for its major benfits:

- High bandwidth and low latency for intensive workloads.
- Scalable design (easy to expand by adding switches).
- Optimized for East–West traffic (server-to-server communication).
- Better load balancing using ECMP.

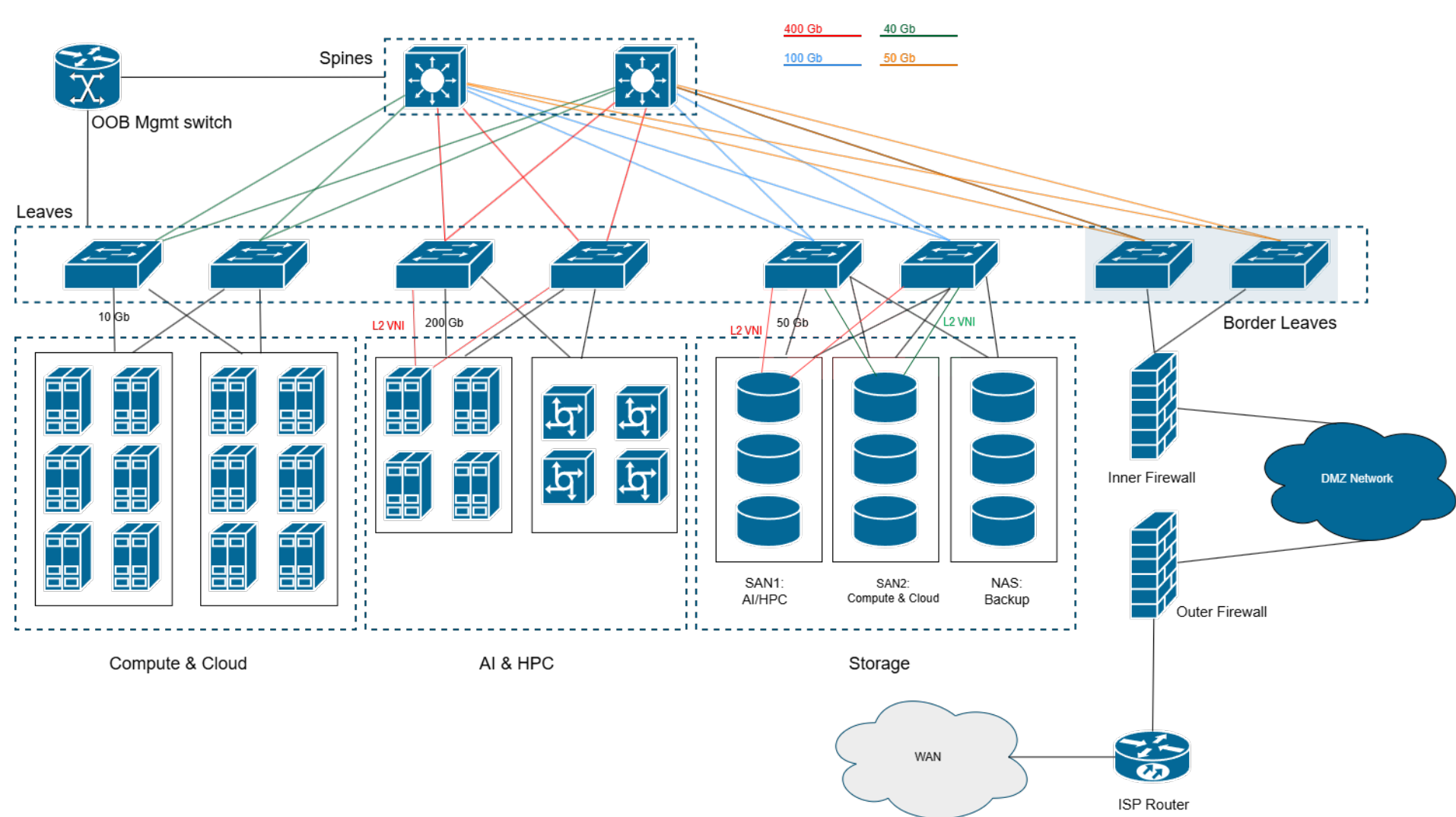


Figure 2. Proposed Solution Architecture Overview

Physical Switch Specifications:

The following table presents the selected switching equipment for the proposed Leaf–Spine architecture, including their roles, configurations, and justification within the data center design.

Device Role	Referenece Model	Port Configuration	Quantity	Main features
Spine Switch	Arista 7060DX5-64S	64× 400G QSFP-DD + 2× SFP+ (upgradable to 800G)	2	Non-blocking 800/400/100/50G fabric, DLB, PFC, RoCEv2
Border Leaf	Arista 7050SX3-48C8C	48x SFP+ 8x QSFP100	2	WAN uplinks, inter-VRF routing, firewall connectivity
General Leaf	Arista 7050TX	48× 10G-T + 6× QSFP+	2	Application servers, cloud compute, user-facing services
AI/HPC Leaf	Arista 7060DX5-64S	64× 400G QSFP-DD + 2× SFP+ (upgradable to 800G)	2	Non-blocking fabric, DLB, PFC, ECN, RoCEv2 (RDMA)
Storage Leaf	Arista 7050X3	8× 200G QSFP28 (upgradable to 400G)	2	NVMe-oF, NFS, S3 traffic
OOB Mgmt Switch	Arista 7010T	48 port 10/100/1000 and 4 port 10GbE	1	IPMI/iDRAC, Advanced Event Management (AEM)

Table 4. Physical switch specifications and references for the proposed architecture.



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Physical Compute Inventory

The following table summarizes the initial compute and storage nodes of the proposed infras-
tructure; additional nodes can be added as needed.

Node Type	Form Factor	CPU & RAM	Storage	Quantity
General Compute	2U	2× Intel Xeon Scalable (16–24c) 256–512 GB DDR5	2× NVMe + SAN	6
AI	4U	2× AMD EPYC 9004 (32c) 512 GB–1 TB 4× NVIDIA A100 80GB	4× NVMe local + NVMe-oF	2
HPC	2U	2× AMD EPYC 9554 (64c) 512 GB–2 TB	2× NVMe + parallel FS	4
Storage	2U	1× Intel Xeon (12c) 128 GB	12× 15TB NVMe + 24× 18TB 3(erasure) HDD	
Management	3U	1× Intel Xeon (8c) 64 GB	2× SSD (RAID-1)	2 (HA)

Table 5. Compute and Storage cluster specifications.

Physical Cabling and Transceiver Plan

- Up to 400G Spine–Leaf uplinks ensure a near non-blocking fabric with 1.23:1 oversubscription.
- AI/HPC nodes use 200G ConnectX RNICs with RoCEv2 (NVMe-oF ready).
- MLAG between Leaf pairs enables active-active high availability.
- Dedicated iDRAC ports for out-of-band management, isolated from data traffic.
- Border Leaf pair connected to the WAN via a secure access for external services with load balancing for BAC operation.

Logical Architecture: Service Domains

The following table presents the logical service domains of the proposed architecture:

Domain	Key Services	Leaf Pair
General Cloud	BAC Orientation, Talents, Moodle, Admin applications, VMs Provision	LEAF-01/02
AI & HPC	GPU training jobs, AI labs, MPI simulations, parallel computing labs	LEAF-03/04
Storage	NVMe-oF block storage	LEAF-05/06
DMZ / Border	Public services, WAN uplinks, firewall connectivity	Border Leaf pair

Table 6. Architecture Logical Service domains

Underlay Design

For the logical Architecture, we choose to work with BGP as underlay protocol, IPv4 prefixes (loopbacks) are exchanged via BGP extended next-hop encoding.

Spines share ASN 65000. Each leaf receives a unique private ASN (65001 65006). Border leaves use ASN 65100.

Of course, ECMP is used, also BFD (Bidirectional Forwarding Detection), where each link uses it to detect failures in less than a second.

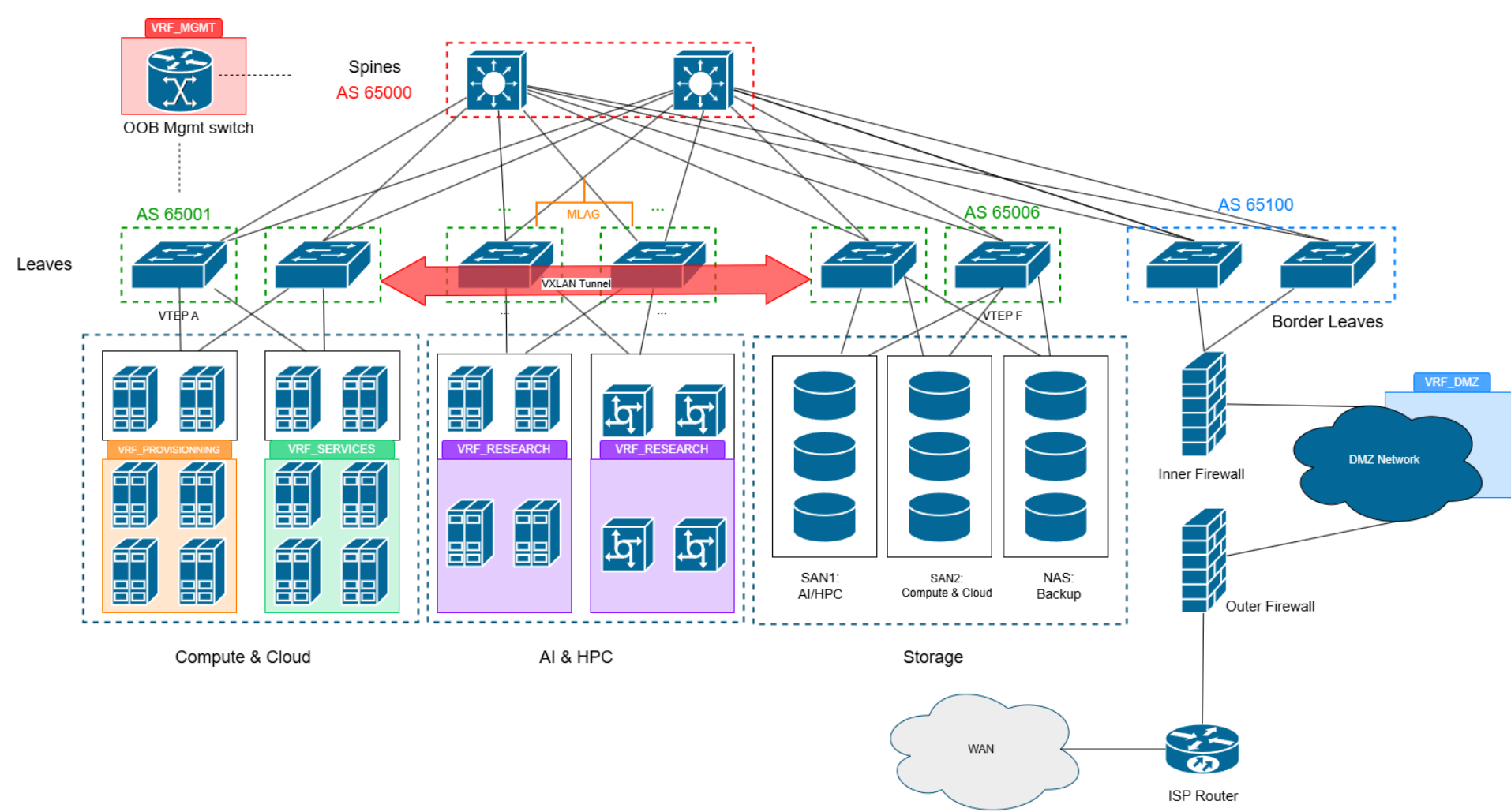


Figure 3. Proposed Logical Solution Architecture

VXLAN/EVPN Overlay Design & VRF Segmentation

EVPN/VXLAN Configuration

- All leaf switches act as VTEPs. MAC/IP routes are advertised via BGP EVPN **Type-2**; L3 prefixes via **Type-5**.
- Symmetric IRB** (Integrated Routing and Bridging) model ensures consistent L2+L3 forwarding at every leaf.
- Each VNI maps 1:1 to a VLAN locally and a VRF L3VNI globally.
- SVIs** (Switched Virtual Interfaces) anchor each VLAN to its parent VRF, enabling seamless VM mobility across the fabric without gateway reconfiguration.
- BUM traffic is handled via **ingress replication** to avoid multicast dependency in the underlay.

VRF-to-L3VNI Mapping:

VRF	L3VNI	Scope & Usage
VRF_MGMT	20002	Admin SSH/RDP, Hypervisor mgmt (proxmox), Monitoring, OOB network devices
VRF_INFRA	10001	Orchestration, VM live migration, PXE Boot provisioning
VRF_SERVICES	40004	AD/LDAP identity, DNS, DHCP, NTP, NAS file access, Backup, Supervision
VRF_RESEARCH	50005	Dedicated network segment for AI, HPC, and research computing
VRF_DMZ	40005	Public-facing application services
VRF_ADMINISTRATION	30003	HR, Payroll, Academic records, Institutional management
VRF_PROVISIONING	60006	VMs (private cloud) — micro-segmented per class/group via Security Groups
SAN1 (AI/HPC)	N/A	L2VNI only — NVMe-oF RoCE storage access for AI/HPC workloads
SAN2 (Compute/Cloud)	N/A	L2VNI only — NVMe-oF/TCP storage access for general compute and cloud workloads

Table 7. VRF-to-L3VNI mapping and functional scope, including storage L2VNIs.

Inter-VRF Routing Policy

The network employs a segmented VRF architecture based on a strict least-privilege model. Crit-
ical shared services (DNS, Auth) are centralized, while sensitive zones (DMZ, Administration) re-
main heavily isolated from (compute, ai/hpc cluster) environments. Cross-boundary access to the
AI/HPC cluster is strictly gated via management hosts, and broad routing is reserved exclusively
for secure management protocols. This design efficiently prevents unauthorized lateral move-
ment between public, educational, and critical infrastructure zones.

Storage Architecture Strategy

The storage design avoids a compromised one-size-fits-all approach by segmenting services into
dedicated silos based on opposing workload constraints:

- AI/HPC (SAN 1):** Prioritizes extreme performance. Requires an isolated, lossless L2 fabric for direct storage-to-GPU RDMA transfers, scaling horizontally by adding NVMe nodes.
- Virtualization (SAN 2):** Prioritizes stability & high availability for VM live migration. Uses standard L2 Ethernet without RDMA overhead, scaling via additional storage arrays.
- General NAS:** Prioritizes broad accessibility. Routed across L3 boundaries (VRFs) for campus-wide file sharing and backups, designed to scale-out with clustered file systems.

Service	Protocol	Connectivity & Scope	Layer
SAN 1 — AI/HPC	NVMe-oF RoCEv2	Pure L2 (VLAN/VNI) on AI Leafs. Lossless fabric (PFC/ECN). Not routed — RDMA-safe.	L2
SAN 2 — Virtualisation	NVMe-oF TCP	Pure L2 (VLAN/VNI) on Virtu Leafs. Standard Ethernet, no loss-less requirement.	L2
NAS — General Use	NFS	L3 routed via VRF_SERVICES. Access filtered through Border Leaf / Firewall policies.	L3

Table 8. Storage tier specifications, protocols, and connectivity model.

QoS Design:

The fabric preserves service quality by design: a 1.23:1 oversubscription ratio and ECMP load
balancing minimize structural congestion. QoS is applied selectively, only where workload behav-
ior specifically demands it.

Class	DSCP	Scheduling / Mechanism	Where Applied / Scope
Network Control	CS6	Strict Priority (policed)	All switches / routing adjacencies, EVPN/BGP, BFD, and control-plane traffic
AI↔AI RDMA	AF41	Dedicated RDMA queue	AI/HPC leaves / AI-to-AI RoCEv2 —sustained elephant flows
HPC↔HPC RDMA	AF31	Dedicated RDMA queue	AI/HPC leaves / HPC-to-HPC RoCEv2 — ultra-low jitter requirement
Storage RDMA	AF21	Dedicated RDMA queue	AI/HPC and Storage leaves/NVMe-oF over RoCEv2 — iso- lated from pause effects
Interactive Access & Control	CS2	High-weight WRR/DWRR	DMZ ingress, border leaves, and service-facing leaves / portals, APIs, bastion SSH, scheduler, authentication
Best-Effort Traffic	BE	Fair-share WRR/DWRR	All in-band fabric switches: NFS, S3, NVMe/TCP, VM and standard east-west traffic

Table 9. Selective QoS class model for the proposed fabric.



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- Ingress leaves mark traffic, DSCP preserved across VXLAN, re-marked at border leaves.
- ECN signals early, DCQCN reduces sender rate, PFC is the last resort, confined to server-facing ports on AI/HPC and Storage leaves, never propagated to the spine.
- Lossless reserved for RDMA flows only, unclassified traffic defaults to best-effort.

General Compute Cluster (Virtualization & Cloud):

The cluster hosts three VM categories: **infrastructure services** (VRF_SERVICES), **DMZ-exposed services**, and **student/researcher labs** (VRF_PROVISIONING), all orchestrated via **proxmox** for native multi-tenancy, self-service provisioning, and automated network isolation over the BGP EVPN/VXLAN fabric.

VM Category	Hosted Services	vCPU	RAM
Infrastructure	DNS, DHCP, LDAP/AD, NTP, Backup, Supervision	~10%	~10%
DMZ / Web	Reverse Proxy, VPN GW, BAC Platform, AI/HPC Portal, SFTP	~25%	~25%
Student & Research	Teaching VMs, Linux envs, project sandboxes	~65%	~65%

Table 10. Static resource distribution across VM workload categories.

Period	BAC Platform	VMs	Mechanism
Academic Semester	Standby only	Full allocation	Default flavor quotas
BAC Orientation Peak	Scaled out (replicas + vCPUs)	Quota-capped	Auto-scaling on load
Exam / Off Period	Moderate (results load)	Minimal	Manual quota policy

Table 11. Dynamic resource reallocation across operational periods.

AI/HPC Cluster:

Defining the optimal architecture for AI and HPC integration required evaluating two approaches: a converged design (AI and HPC nodes under the same leaf) versus dedicated silos (separate leaves for each). Based on ESI's current needs and expected workload scale, the converged approach is the most suitable.

- Initial cluster size is small (few AI and HPC nodes), making dedicated high-speed leaves inefficient and costly due to the resulting underutilization and port waste.
- Provides flexibility to evolve toward dedicated clusters as demand grows

AI & HPC Cluster Overview Compute Jobs are submitted via login nodes to an orchestrator (Kubernetes), which schedules GPU/HPC workloads across compute nodes over a high-bandwidth, lossless fabric.

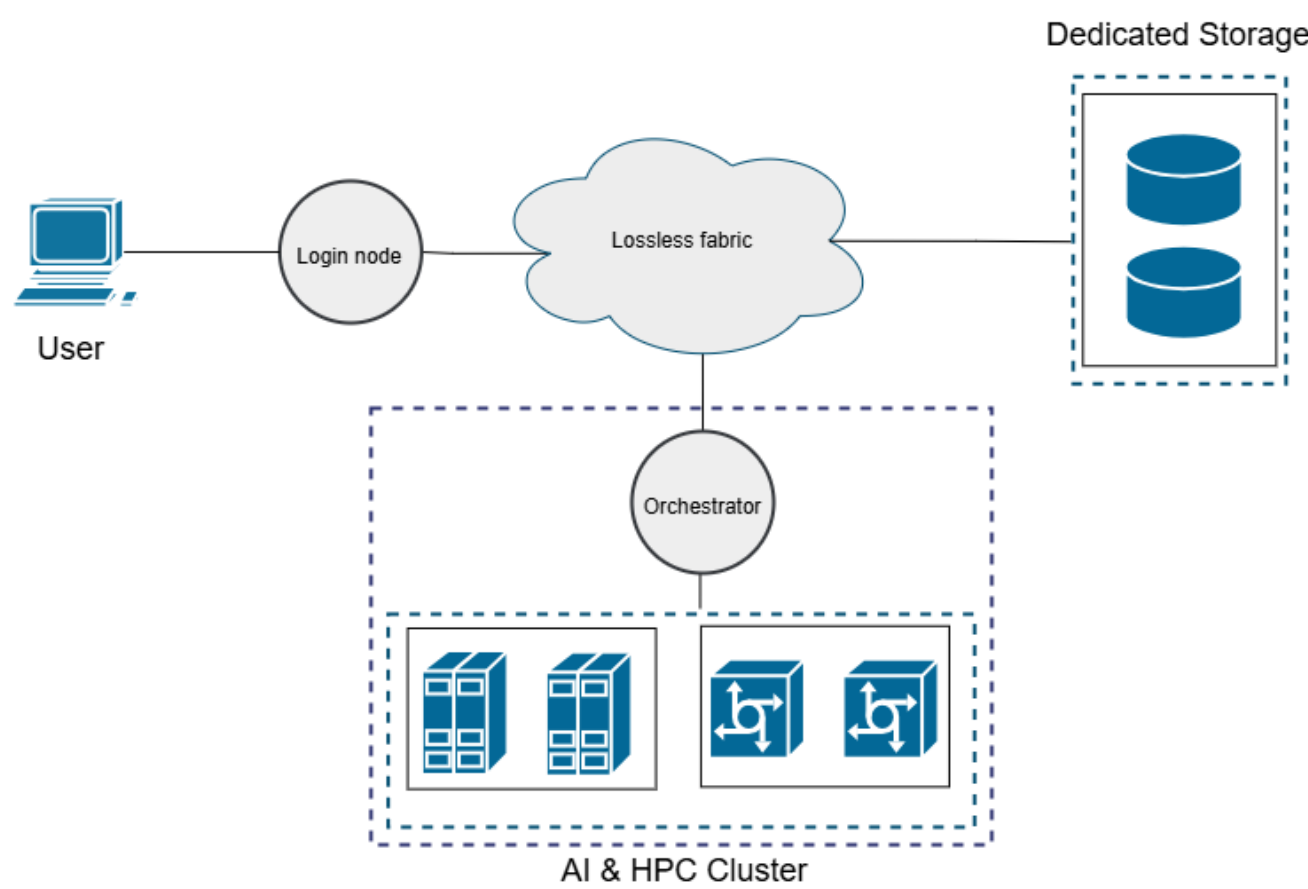


Figure 4. AI & HPC cluster

Security & Access Control

The design adopts a **DMZ-based access model** with strict VRF segmentation across all service domains. No direct Internet-to-internal access is permitted — all inbound traffic terminates in the DMZ, and only explicitly authorized flows are forwarded toward the target internal VRF.

Security & Access:

- **Dual-firewall, dual-vendor boundary:** the outer firewall handles perimeter filtering; the inner firewall enforces a default-deny policy toward all internal zones — vendor diversity eliminates single points of compromise.
- **Authenticated access:** VPN for remote users, RBAC and MFA for administrative operations, all sessions encrypted via TLS/HTTPS/SFTP.
- **Zone isolation:** VRF segmentation separates management, infrastructure, services, research, administration, and private cloud — cross-zone traffic is always firewall-enforced, never bypassed.

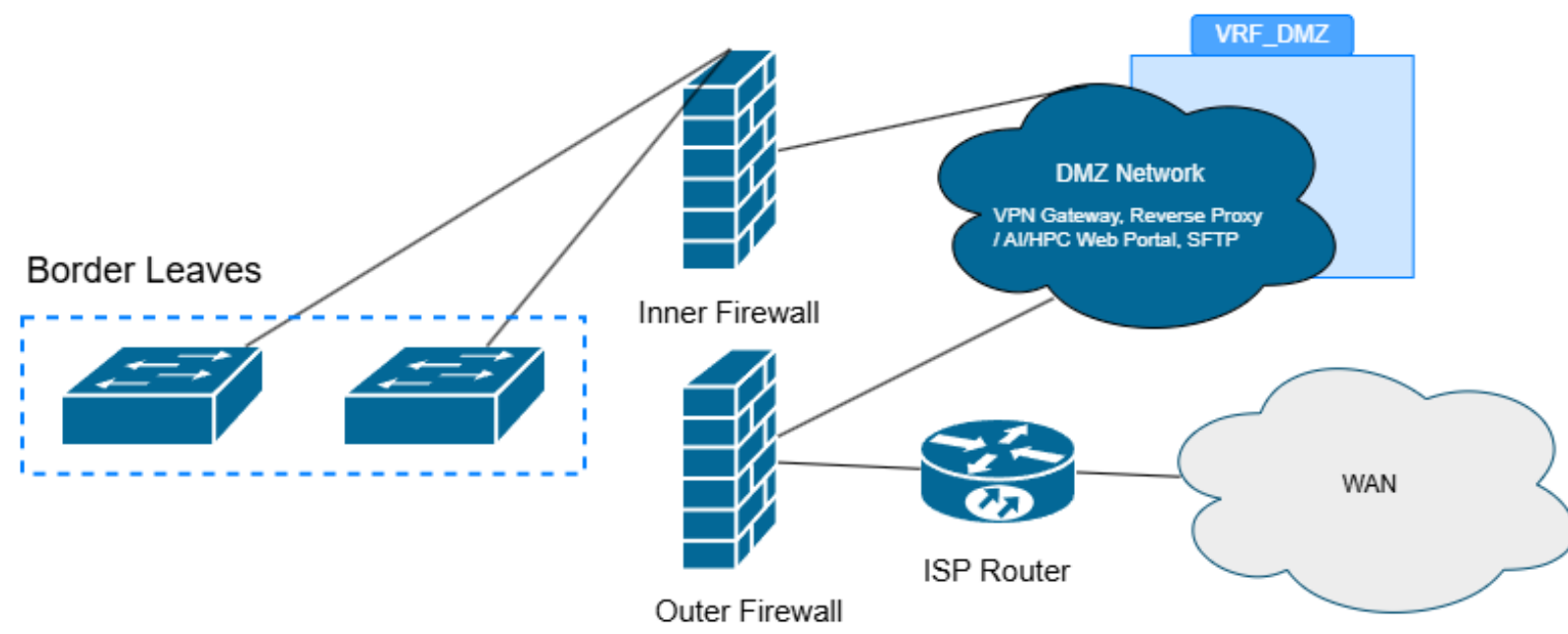


Figure 5. DMZ-based secure access and dual-firewall architecture

Service	Main Role	Authorized Scope
VPN way	Gate- Secure remote access for autho- sized users	Approved internal services through firewall policy
Reverse Proxy / API Gateway	Single public entry point for web & APIs	Approved HTTPS/API traffic to backend only
AI/HPC Web Portal	Job submission and monitoring for AI & HPC workloads	AI/HPC platforms and schedulers only
Bastion Host	Administrative jump point	Restricted SSH/API toward internal systems
SFTP / FTPS	Secure file transfer	Approved storage and dataset access only

Table 12. Key services exposed in the DMZ and their controlled access scope.

Scalability

Two main phases can be scaled to meet the growing demands:

Phase 1: Compute & Service Scaling

Add compute nodes to expand research and final-year project capacity. Nodes connect to existing Leaf pairs, with Ansible automating deployment and BGP EVPN ensuring VM/Container mobility without IP changes, providing zero downtime.

Phase 2: Fabric Scaling

Adding a third spine increases fabric bandwidth by 50%. The fabric is upgradable to 800G to support high-capacity GPUs (Nvidia H100 and B200) over more than 20 year lifecycle, ensuring readiness for interconnection with other data centers and eventually a national AI network.

Centralized Observability & Monitoring

To resolve the lack of centralized system and environmental visibility, we introduce a unified observability stack:

Domain	Technology	Features & Justification
Telemetry		
Network & OS	Prometheus Grafana Labs	Aggregates real-time streaming telemetry from Arista switches and compute nodes into single-pane-of-glass dashboards.
OoB Management		
Environmental	ZABBIX Zabbix	Interfaces with server IPMI/iDRAC over the dedicated 7010T switch to track thermals and hardware health independently.

Table 13. Proposed monitoring stack.

Conclusion:

The proposed architecture provides ESI with a coherent, scalable, and security-enforced modern-ization strategy. It replaces the limitations of the current three-tier environment with a leaf-spine EVPN/VXLAN fabric designed for high bandwidth, low latency, and controlled growth, while strict VRF segmentation, a DMZ-based dual-firewall model, and selective QoS ensure strong isolation, secure access, and predictable service behavior. Combined with dedicated support for AI/HPC workloads, adapted storage paths, and centralized observability, the design answers both present operational needs and future expansion requirements.

In this sense, the project does not only solve current weaknesses; it defines a credible infrastructure foundation for the next generation of ESI services.

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